

RISHITA RAMOLA

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EDUCATION

Graphic Era Deemed To Be University, Dehradun Expected June 2028
B.Tech in Computer Science & Engineering | CGPA: 8.51 / 10

Carmel School, Chamba (ISC) 2024
Class XII: PCM + Computer Science | 85%

Carmel School, Chamba (ICSE) 2022
Class X: 88%

SUMMARY & FIELDS OF INTEREST

- B.Tech CSE student with hands-on experience building **multi-agent LLM systems**, NLP pipelines, and ML-driven applications. Passionate about solving hard NLP problems — from document understanding and natural language query parsing to agentic workflow orchestration. A quick learner who thrives in fast-paced, collaborative environments and is dedicated to staying at the cutting edge of AI/ML research.
- Interests:** Natural Language Processing, LLM-based Agents & Code Generation, Document Understanding, Retrieval-Augmented Generation (RAG), Evaluation Pipelines, ML Foundations, Prompt Engineering.

TECHNICAL SKILLS

- Languages:** Python, C++, Java, C, JavaScript, SQL.
- NLP / ML:** PyTorch, Hugging Face Transformers, Scikit-learn, Pandas, NumPy, LLM Prompt Engineering, Multi-Agent LLM Systems, Agentic Pipelines, Evaluation Frameworks.
- Backend & Cloud:** FastAPI, Flask, Node.js, MongoDB, REST APIs, Docker, AWS (AI Practitioner Certified), Hugging Face Spaces, Gradio, Streamlit.
- Dev Tools:** Git/GitHub, Linux/Unix, CI/CD Pipelines, Jupyter Notebooks.

ACHIEVEMENTS & HACKATHONS

- Meta PyTorch Conference — National Finalist:** Selected among top teams nationwide for ML/NLP research; presented at the **finals in Bangalore**.
- State Topper — International Olympiad of Mathematics:** Ranked #1 in state for logical reasoning & analytical problem-solving.
- 1st Prize — Debate (GEU):**
- INSPIRE Award — Dept. of Science & Technology, Govt. of India:** National recognition for innovation and scientific aptitude.
- Hack-o-Holic — 24-Hour Hackathon:** Built and deployed a full-stack ML-powered predictive system within a 24-hour competitive sprint.

PROJECTS

Orator — Speech Mastery Platform — *Solo Project | Vite, Vanilla JS, Web Audio/Speech APIs, CSS | In Progress* May 2026

- Engineered a high-performance, language-agnostic speech analysis platform using **Web Audio API** for real-time acoustic analysis (FFT pitch detection) and **Web Speech API** for STT and metric tracking.
- Developed a custom **Emotion & Intent Layer** for paralinguistic analysis (EAS scoring) and an **Adaptive Learning Path** leveraging spaced repetition for phoneme mastery and L1 interference prediction.
- Designed a fully client-side, zero-backend architecture featuring a premium dark-themed UI, dynamic 60fps Canvas-based waveform visualizations, and a cross-cultural register coach.

Judicial Multi-Agent Debate Arena — *Team Lead | Python, Groq LLMs, Hugging Face, Docker | In Progress* Apr 2026

- Architected a **3-agent LLM pipeline** (Plaintiff, Defendant, Judge) with structured prompt chains performing IRAC-based legal reasoning — a complex **agentic workflow** for document understanding and context synthesis.
- Designed **anti-hallucination guardrails** with live legal database verification and citation accuracy checks — functioning as a custom **evaluation framework** for LLM output quality.
- Engineered natural language query parsing to extract legal entities, statutes, and arguments from unstructured case descriptions; deployed on **Hugging Face Spaces** with Gradio UI.

Echoes of the Sentinel — 2D Horror Game — *Team Lead | Java, AWT/Swing, OOP Patterns* Mar 2026

- Led design of a modular game engine using entity-component systems and OOP patterns (State, Observer, Factory); built custom rendering pipeline and time-decay scoring.

Digital Twin of You — Predictive Behavioral Engine — *Team Lead | React, FastAPI, Scikit-learn | In Progress* 2026

- Built a **Decision Tree classification pipeline** (Scikit-learn) with custom feature engineering using Pandas/NumPy to predict behavioral outcomes from 14-day time-series data.
- Designed observation and intervention engines performing real-time pattern detection and chain analysis; React frontend + FastAPI backend + SQLite.

Smart Ambulance Routing System — *Team Lead | JavaScript, Leaflet.js, Node.js* Apr 2026

- Led a team building a **real-time emergency dispatch system** optimizing ambulance→patient→hospital routing with live geolocation data.
- Designed high-concurrency APIs for mission-critical data integrity; implemented dual-path animated visualization on an interactive Leaflet map.

GreenMart — Supply Chain Optimizer — *Team Lead | C++, Node.js* 2025

- Memory-safe C++ backend with Max-Heap delivery optimization; Node.js middleware for real-time buyer workflows.

LEADERSHIP & EXTRACURRICULARS

- School Captain:** Elected to lead a student body of 1,000+; managed a 12-member council and coordinated major institutional logistics.
- Strengths:** Collaborative Teamwork, Troubleshooting, Technical Documentation, Leadership, Analytical Problem Solving, Strategic Time Management.
- Competitive Coding:** Active participant in Code Hunt and Graph-e-Thon; focus on Graph theory and Algorithms.
- Technical Argumentation & Global Debating.
- Passionate about Singing and Dancing.